

Template Title

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Contents

1	Question 1	1
2	Question 2	2
3	Question 3	3

1 Question 1

1. Consider the following action (as usual, $g = \det g_{\mu\nu}$)

$$S = \frac{1}{16\pi G_N} \int d^n x \sqrt{-g} (R - 2\Lambda) \quad (1)$$

- (a) Derive the equations of motion. You can use $\delta R = R_{\mu\nu} \delta g^{\mu\nu}$ up to a total derivative. Check that $\Lambda = 0$ case reproduces Einstein's equations in the vacuum.

Figure 1.1

- (c) A rocket which stays stationary at radius R_1 and $\theta = \pi/2, \phi = 0$, sends a radio signal of frequency w_1 to an observer which moves inertially (without any forces acting on it) along the stable circular orbit of radius R_2 in the $\theta = \pi/2$ plane. The signal propagates in the radial direction (and hence has $\theta = \pi/2, \phi = 0$). What is the frequency of the signal as received by the observer?

Figure 1.2

- (b) [10 points] Compute the temperature of this black hole and verify the first law of thermodynamics.

Figure 1.3

2 Question 2

- (a) Consider a **timelike** geodesic which has $\theta = \pi/2, \phi = \pi/2$. Use the Killing vectors, ∂_t and ∂_ψ to write two conserved quantities: energy E and angular momentum L . Use expressions for E, L from to write the geodesics equation in the form

$$\frac{1}{2} \left(\frac{dr}{d\lambda} \right)^2 + V(r) = \frac{1}{2} E^2$$

Calculate the effective potential $V(r)$.

Figure 2.1

- (b) Compute the radius R of a circular ($r = \text{const}$) orbit. Is it a stable or unstable circular orbit? What are the smallest and the largest values of R ?

Figure 2.2

[10 points] At some point in spacetime, the astronaut flies near a stationary observer who is located at radius r_* . Use the result of (a) to compute the **speed** of the astronaut as measured by the observer.

Figure 2.3

3 Question 3

3. (a) Consider the FRW Universe described by the metric

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right] \quad (5)$$

The two Friedmann equations with the cosmological constant Λ are

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P) + \frac{\Lambda}{3}$$

and

$$\frac{\ddot{a}}{a} + \frac{2\dot{a}^2}{a^2} + \frac{2k}{a^2} = 4\pi G(\rho - P) + \Lambda$$

where P is the pressure and ρ is the density of the perfect fluid that fills the FRW universe. Dot denotes differentiation with respect to time t . Assume $\Lambda > 0$. How can one determine whether the universe is closed (3-sphere), open (hyperboloid), or flat, given the density ρ , cosmological constant Λ and the Hubble parameter $H = \dot{a}/a$ at a given time?

Figure 3.1

- Consider a universe filled with matter whose equation of state is $P = \omega\rho$. From the conservation of stress energy tensor and $\Gamma_{01}^1 = \Gamma_{02}^2 = \Gamma_{03}^3 = \dot{a}/a$, $\Gamma_{00}^\mu = \Gamma_{20}^1 = 0$ show that $\rho \simeq a^{-\alpha}$ and compute the value of α .

Figure 3.2

- (c) Consider now **radiation** filled spherical universe with nonvanishing Λ . Find a *static* (independent of t) solution of Friedmann equations above and relate the size of the Universe, a , to the value of the cosmological constant Λ .

Figure 3.3